

Aerobic oxidation of indole carbinols using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/NaCl as catalysts†Cite this: *Org. Biomol. Chem.*, 2013, **11**, 4186Jinxian Liu^a and Shengming Ma^{*a,b}

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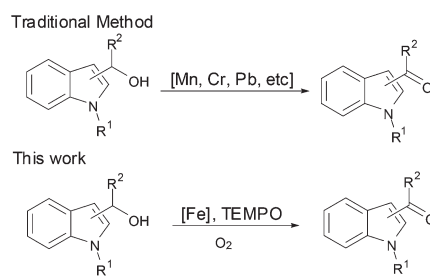
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A practical aerobic oxidation of indole carbinols using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/NaCl in DCE at room temperature and atmospheric pressure of oxygen affording aldehydes or ketones in good to excellent yields was developed. Furthermore, when using the industrially favored solvent toluene instead of DCE and air instead of pure oxygen, this protocol also works smoothly, demonstrating its high potential for possible industrial application.

Introduction

Carbonyl compounds with indole units are important in organic synthesis since a large number of indole carbonyls with bioactivity have been found in natural products.¹ Traditionally, such compounds could be obtained by oxidation of the corresponding alcohols using at least a stoichiometric amount of oxidants such as MnO_2 ,² CrO_3 ,³ $\text{Pb}(\text{OAc})_4$,⁴ DMSO,⁵ hypervalent iodine compounds,⁶ *etc.*⁷ (Scheme 1). However, from an economical and environmental viewpoint, such protocol would produce an almost equivalent amount of oxidant-derived waste, and place a heavy burden on the environment. Thus, development of a novel eco-friendly and economic approach for such a transformation is highly desired.

Molecular oxygen is a clean and cheap oxidant, and releases water as the sole byproduct, thus it's an ideal terminal oxidant, especially for industrial applications. In the last three decades, numerous aerobic oxidation protocols with metal catalysts or even without a metallic catalyst have been reported.⁸ 2,2,6,6-Tetramethylpiperidine-*N*-oxyl (TEMPO) has been used as a catalyst to oxidize alcohols with various oxidants including molecular oxygen.^{8e,9} As we know, there are very limited reports for the oxidation of indole carbinols using molecular oxygen as terminal oxidant in the literature.¹⁰ Recently, this group has developed a general and practical approach to aerobic oxidation of alcohols bearing a wide range of substrates under mild conditions using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/NaCl as catalyst, yielding aldehydes or ketones in good to



Scheme 1 Oxidation of indole carbinols.

excellent yields.⁵ Therefore, we reasoned that such a catalytic protocol could also be applied to indole carbinols, providing an efficient alternative.

Results and discussion

We tried to optimize the reaction conditions with regard to additives, nitrates, solvents and catalyst loading; the results are summarized in Table 1. A commonly used kitchen chemical, we found that sodium chloride is an important accelerator for the $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO-catalyzed aerobic oxidation of alcohols.¹¹ Thus, we first carried out a control experiment to investigate the effect of chloride ion on this oxidation (Table 1, entries 1 and 2). We found that when 10 mol% NaCl was added, the catalytic efficiency was greatly improved, and the reaction time was reduced from 15 hours to 2 hours. The role of Cl^- in this reaction is not quite clear yet; it probably works as a ligand to Fe^{3+} to accelerate the oxidation.¹¹ Copper nitrate could also catalyze this reaction; however, the reaction is slow (6 h) (Table 1, entry 3). Therefore, we still chose iron nitrate as the co-catalyst. We next examined the effect of solvent on the aerobic oxidation reaction at room temperature (Table 1, entries 4 and 5) and found that this reaction could be carried out in ethyl acetate, THF or DCE due to its high reactivity. For

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Table 1 Optimization of the aerobic oxidation of (1-tosyl-1*H*-indol-2-yl)-carbinol^a

Entry	Solvent	Time (h)	Yield of 2a (%)
1	DCE	2	99
2 ^b	DCE	15	93
3 ^c	DCE	6	95
4	Ethyl acetate	3	97
5	THF	5	95
6 ^d	DCE	4	96
7 ^e	DCE	43	93

^a Standard reaction conditions: 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl in 4 mL of DCE. ^b Without the addition of 10 mol% NaCl. ^c Cu(NO₃)₂·3H₂O was used instead of Fe(NO₃)₃·9H₂O. ^d The reaction was conducted with 5 mol% Fe(NO₃)₃·9H₂O, 5 mol% TEMPO, and 5 mol% NaCl in 4 mL of DCE. ^e The reaction was conducted with 2 mol% Fe(NO₃)₃·9H₂O, 2 mol% TEMPO, and 2 mol% NaCl in 4 mL of DCE.

the consideration of solubility and efficiency, we chose DCE as solvent in this reaction. We further investigated the catalyst loading (Table 1, entries 6 and 7) and found that when 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl were used, the reaction could be rapidly completed in 2 hours with an almost quantitative yield; however, when the catalyst loading of each catalyst was reduced to 5 mol%, the reaction time doubled; when further reducing the catalyst loading of Fe(NO₃)₃·9H₂O, TEMPO, and NaCl to 2 mol%, the reaction time was prolonged to 43 hours.

Therefore, we chose 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl in DCE as standard reaction conditions for further investigation into the substrate scope of this reaction, with the results being summarized in Table 2. This protocol could not only be applied to the primary indole-2-carbinols (Table 2, entries 1–3) but also to the corresponding secondary alcohols (Table 2, entries 4 and 5) affording aldehydes and ketones, respectively, with excellent yields. The

Table 2 Room temperature aerobic oxidation of indole-2-carbinols to corresponding aldehydes or ketones^a

Entry	Substrate	R ¹	R ²	Time (h)	Yield of 2 (%)
1	1a	H	H	2	99 (2a)
2	1b	H	5-F	3	90 (2b)
3	1c	H	5-COOMe	12	84 (2c)
4	1d	CH ₃	H	14	84 (2d)
5	1e	<i>n</i> -C ₅ H ₁₁	H	24	92 (2e)

^a Standard reaction conditions: 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl in 4 mL of DCE.

indole ring could bear electron-withdrawing groups such as F and COOMe (Table 2, entries 2 and 3).

N-Ts, Boc or Ac are required for this transformation (Table 3, entries 1–3). When R¹ = Ts or Ac, the substrate could be oxidized to corresponding aldehyde rapidly with good yield; when R¹ = Boc, 87% isolated yield could be obtained within 5 hours. The R² group could be aryl, alkyl, a terminal C–C triple bond or an allene (Table 3, entries 4–7).

Table 3 Aerobic oxidation of indole-3-carbinols to corresponding aldehydes or ketones – the effect of *N*-protecting group^a

Entry	Substrate	R ¹	R ²	Time (h)	Yield of 2 (%)
1	1f	Ts	H	1	80 (2f)
2	1g	Boc	H	5	83 (2g)
3	1h	Ac	H	1.5	87 (2h)
4	1i	Ac	Ph	5	89 (2i)
5	1j	Ac	C ₂ H ₅	4.5	89 (2j)
6	1k	Ac	HC≡C	9	94 (2k)
7	1l	Ac	CH=C=CH ₂	5	69 (2l)

^a Standard reaction conditions: 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl in 4 mL of DCE.

We further investigated the usability of air instead of pure oxygen; the results are summarized in Table 4. The transformation with air could be completed within 3 hours with 97% yield (Table 4, entry 1). When reducing the catalyst loading to 5 mol% each of Fe(NO₃)₃·9H₂O, TEMPO, and NaCl, the reaction time doubled (Table 4, entry 2); when changing the reaction solvent to toluene,¹² the reaction could also be completed in 10 hours (Table 4, entry 3).

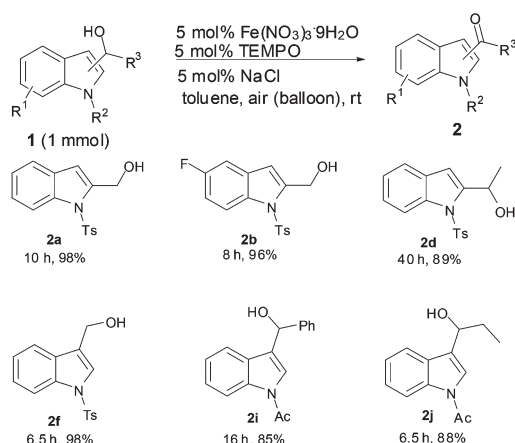
Table 4 Optimization of the aerobic oxidation of (1-tosyl-1*H*-indol-2-yl)-carbinol using air instead of oxygen^a

Entry	<i>x</i>	Solvent	Time (h)	Yield of 2a (%)	Yield of 2a (%), NMR
1	10	DCE	3	97	
2	5	DCE	8	96	
3	10	Toluene	2	99	
4	5	Toluene	10	99	

^a Standard reaction conditions: 10 mol% Fe(NO₃)₃·9H₂O, 10 mol% TEMPO, and 10 mol% NaCl in 4 mL of DCE.

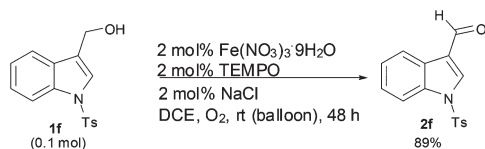
We then applied these optimized reaction conditions using air as oxidant and toluene as solvent to some indole carbinols;

the results are summarized in Scheme 2. We found that both indole-2-carbinols and indole-3-carbinols could be transformed to corresponding carbonyl compounds in excellent yield under the standard conditions.



Scheme 2 Room temperature aerobic oxidation of indole carbinols in toluene using air as oxidant.

To further show the practicality and efficiency of this catalytic system, a 0.1 mol reaction of (1-tosyl-1H-indol-3-yl)carbinol was conducted by using 2 mol% each of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO and NaCl to give 1-tosyl-1H-indole-3-carbaldehyde in 89% isolated yield in 48 h by simple recrystallization.



Conclusions

In summary, we have developed a practical aerobic oxidation protocol for various types of indole carbinols using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/NaCl as catalysts under mild conditions, affording the corresponding indole carbonyl compounds in good to excellent yields. This eco-friendly and economically oxidation system may be applied in academic laboratories as well as industrial scale production.

Experimental section

General experimental methods

^1H , ^{13}C and ^{19}F NMR spectra were recorded with a Bruker AN 300 MHz spectrometer. ^1H NMR spectra (300 MHz) were recorded using TMS as an internal standard (δ 0 ppm). ^{13}C NMR spectra (75 MHz) were recorded using CDCl_3 as an internal standard (δ 77 ppm). ^{19}F NMR spectra (282 MHz) were recorded using CFCl_3 as an internal standard (δ 0 ppm). Infrared spectra were recorded on an FTIR spectrometer. Mass

spectra were recorded in EI mode. HRMS spectra were recorded in EI mode.

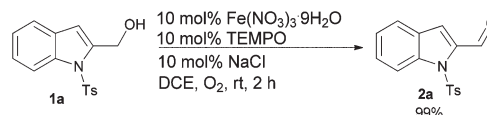
General procedure for the synthesis of aldehydes or ketones using molecular oxygen as oxidant (Method A)

To a 10 mL three-necked flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol), TEMPO (0.1 mmol), NaCl (0.1 mmol) and DCE (4 mL). Alcohol (1 mmol) was then added to the suspension and the resulting mixture was placed under an atmosphere of oxygen from a balloon and stirred at room temperature until the reaction was complete, as monitored by TLC (eluent: petroleum ether–ethyl acetate = 3 : 1). The resulting mixture was purified by column chromatography on silica gel (petroleum ether–ethyl acetate = 10/1) to afford the corresponding aldehyde or ketone.

General procedure for the synthesis of aldehydes or ketones using air as oxidant (Method B)

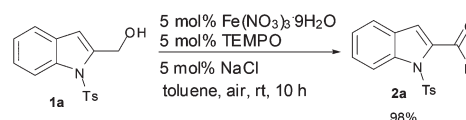
To a 10 mL three-necked flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.05 mmol), TEMPO (0.05 mmol), NaCl (0.05 mmol) and toluene (4 mL). Alcohol (1 mmol) was then added to the suspension and the resulting mixture was stirred at room temperature with an air balloon until the reaction was complete, as monitored by TLC (eluent: petroleum ether–ethyl acetate = 3 : 1). The resulting mixture was purified by column chromatography on silica gel (petroleum ether–ethyl acetate = 10/1) to afford the corresponding aldehyde or ketone.

1. Synthesis of 1-tosyl-1H-indole-2-carbaldehyde (2a).



Method A: the reaction of (1-tosyl-1H-indol-2-yl)carbinol (301.3 mg, 1 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (40.5 mg, 0.1 mmol), TEMPO (15.6 mg, 0.1 mmol), and NaCl (5.8 mg, 0.1 mmol) in DCE (4 mL) afforded 1-tosyl-1H-indole-2-carbaldehyde¹³ (296.3 mg, 99%) as a white solid: m.p. 138–139 °C (hexane–ethyl acetate).

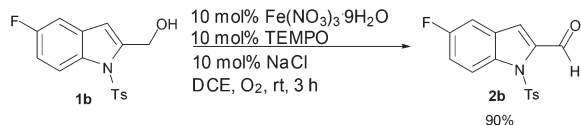
^1H NMR (300 MHz, CDCl_3) δ 10.54 (s, 1H, CHO), 8.24 (d, J = 8.4 Hz, 1H, Ar-H), 7.66 (d, J = 8.4 Hz, 2H, Ar-H), 7.62 (d, J = 8.4 Hz, 1H, Ar-H), 7.52 (t, J = 7.5 Hz, 1H, Ar-H), 7.47 (s, 1H, Ar-H), 7.31 (t, J = 7.5 Hz, 1H, Ar-H), 7.19 (d, J = 8.4 Hz, 2H, Ar-H), 2.33 (s, 3H, CH_3); ^{13}C NMR (75.4 MHz, CDCl_3) δ 183.3, 145.6, 138.5, 137.8, 134.6, 130.0, 128.8, 128.2, 126.6, 124.8, 123.6, 118.8, 115.4, 21.6; IR (neat) 2928, 1673, 1595, 1529, 1441, 1409, 1364, 1346, 1229, 1172, 1148, 1120, 1088, 1041 cm^{-1} ; MS(EI) m/z (%) 299 (M^+ , 30.92), 91 (100).



Method B: the reaction of (1-tosyl-1H-indol-2-yl)carbinol (300.7 mg, 1 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.4 mg, 0.05 mmol),

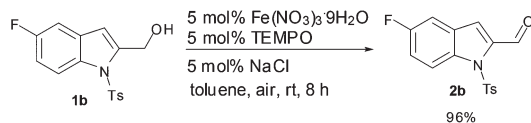
TEMPO (8.0 mg, 0.05 mmol), and NaCl (3.0 mg, 0.05 mmol) in toluene (4 mL) afforded 1-tosyl-1*H*-indole-2-carbaldehyde (291.7 mg, 98%) as a white solid.

2. Synthesis of 5-fluoro-1-tosyl-1*H*-indole-2-carbaldehyde (2b).



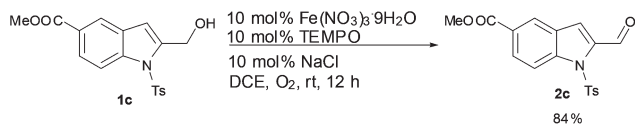
Method A: the reaction of (5-fluoro-1-tosyl-1*H*-indol-2-yl)carbinol (319.8 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.6 mg, 0.1 mmol), TEMPO (15.6 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded 5-fluoro-1-tosyl-1*H*-indole-2-carbaldehyde (286.7 mg, 90%) as a white solid: m.p. 152–153 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 10.53 (s, 1H, CHO), 8.20 (dd, *J*₁ = 9.9 Hz, *J*₂ = 4.2 Hz, 1H, Ar-H), 7.63 (d, *J* = 8.4 Hz, 2H, Ar-H), 7.41 (s, 1H, Ar-H), 7.30–7.18 (m, 4H, Ar-H), 2.34 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 183.2, 160.0 (d, *J* = 238.0 Hz), 145.8, 138.9, 134.7, 134.3, 130.0, 129.1 (d, *J* = 10.4 Hz), 126.6, 118.1 (d, *J* = 4.4 Hz), 117.2 (d, *J* = 25.8 Hz), 116.8 (d, *J* = 9.3 Hz), 108.5 (d, *J* = 23.7 Hz), 21.6; ¹⁹F NMR (282 MHz, CDCl₃) δ –117.6; IR (neat) 2928, 1676, 1595, 1529, 1443, 1411, 1362, 1344, 1305, 1239, 1165, 1121, 1086, 1045 cm^{–1}; MS(EI) *m/z* (%) 317 (M⁺, 26.44), 91 (100); Anal. calcd for C₁₆H₁₂FN₂O₃S: C 60.56; H 3.81; N 4.41; Found: C 60.49; H 3.81; N 4.22.



Method B: the reaction of (5-fluoro-1-tosyl-1*H*-indol-2-yl)carbinol (319.6 mg, 1 mmol), Fe(NO₃)₃·9H₂O (20.8 mg, 0.05 mmol), TEMPO (7.9 mg, 0.1 mmol), and NaCl (2.7 mg, 0.1 mmol) in toluene (4 mL) afforded 5-fluoro-1-tosyl-1*H*-indole-2-carbaldehyde (304.6 mg, 96%) as white solid.

3. Synthesis of methyl 2-formyl-1-tosyl-1*H*-indole-5-carboxylate (2c).

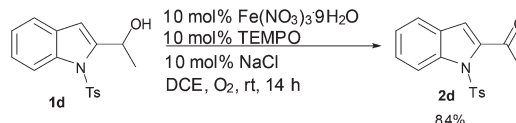


Method A: the reaction of methyl 2-(hydroxymethyl)-1-tosyl-1*H*-indole-5-carboxylate (357.5 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.3 mg, 0.1 mmol), TEMPO (15.8 mg, 0.1 mmol), and NaCl (5.7 mg, 0.1 mmol) in DCE (4 mL) afforded methyl 2-formyl-1-tosyl-1*H*-indole-5-carboxylate (298.0 mg, 84%) as a white solid: m.p. 139–140 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 10.53 (s, 1H, CHO), 8.34 (s, 1H, Ar-H), 8.27 (d, *J* = 8.7 Hz, 1H, Ar-H), 8.17 (dt, *J*₁ = 8.8 Hz, *J*₂ = 1.4 Hz, 1H, Ar-H), 7.67 (d, *J* = 8.4 Hz, 2H, Ar-H), 7.50 (s, 1H, Ar-H), 7.21 (d, *J* = 8.4 Hz, 2H, Ar-H), 3.93 (s, 3H, OCH₃), 2.33 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 182.9, 166.4, 146.0,

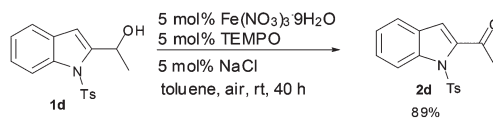
140.6, 138.7, 134.4, 130.1, 129.4, 127.8, 126.9, 126.7, 125.9, 118.6, 115.1, 52.3, 21.6; IR (neat) 2988, 2901, 1716, 1669, 1614, 1596, 1534, 1462, 1436, 1409, 1374, 1307, 1286, 1256, 1233, 1162, 1137, 1086, 1050 cm^{–1}; MS(EI) *m/z* (%) 357 (M⁺, 27.95), 91 (100); Anal. calcd for C₁₈H₁₅NO₅S: C 60.49; H 4.23; N 3.92; Found: C 60.29; H 4.15; N 3.64.

4. Synthesis of 1-(1-tosyl-1*H*-indol-2-yl)ethanone (2d).



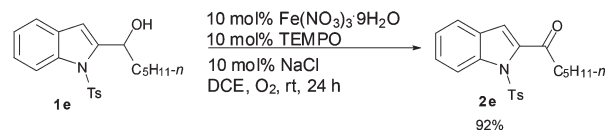
Method A: the reaction of 1-(1-tosyl-1*H*-indol-2-yl)ethanol (315.9 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.1 mg, 0.1 mmol), TEMPO (15.7 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(1-tosyl-1*H*-indol-2-yl)ethanone (263.7 mg, 84%) as a white solid: m.p. 130–131 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.15 (d, *J* = 8.4 Hz, 1H, Ar-H), 7.82 (d, *J* = 8.1 Hz, 2H, Ar-H), 7.55 (d, *J* = 7.8 Hz, 1H, Ar-H), 7.44 (t, *J* = 7.8 Hz, 1H, Ar-H), 7.30–7.18 (m, 3H, Ar-H), 7.10 (s, 1H, Ar-H), 2.63 (s, 3H, COCH₃), 2.33 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 191.7, 144.8, 139.9, 138.8, 135.1, 129.4, 128.3, 127.5, 127.3, 124.3, 122.8, 117.3, 115.7, 29.6, 21.5; IR (neat) 1689, 1537, 1442, 1360, 1330, 1312, 1266, 1223, 1169, 1147, 1125, 1102, 1089, 1018, 969 cm^{–1}; MS(EI) *m/z* (%) 313 (M⁺, 25.73), 91 (100); Anal. calcd for C₁₇H₁₅NO₃S: C 65.16; H 4.82; N 4.47; Found: C 65.09; H 4.78; N 4.17.



Method B: the reaction of 1-(1-tosyl-1*H*-indol-2-yl)ethanol (315.2 mg, 1 mmol), Fe(NO₃)₃·9H₂O (20.0 mg, 0.05 mmol), TEMPO (7.9 mg, 0.05 mmol), and NaCl (3.0 mg, 0.05 mmol) in toluene (4 mL) afforded 1-(1-tosyl-1*H*-indol-2-yl)ethanone (278.2 mg, 89%) as a white solid.

5. Synthesis of 1-(1-tosyl-1*H*-indol-2-yl)hexan-1-one (2e).

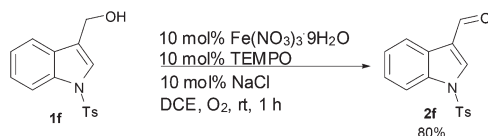


Method A: the reaction of 1-(1-tosyl-1*H*-indol-2-yl)hexan-1-ol (370.1 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.1 mg, 0.1 mmol), TEMPO (15.6 mg, 0.1 mmol), and NaCl (5.6 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(1-tosyl-1*H*-indol-2-yl)hexan-1-one (340.2 mg, 92%) as a white solid: m.p. 97–98 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.12 (d, *J* = 8.4 Hz, 1H, Ar-H), 7.85 (d, *J* = 8.4 Hz, 2H, Ar-H), 7.54 (d, *J* = 7.8 Hz, 1H, Ar-H), 7.43 (t, *J* = 7.8 Hz, 1H, Ar-H), 7.31–7.20 (m, 3H, Ar-H), 7.03 (s, 1H, Ar-H), 2.96 (t, *J* = 7.4 Hz, 2H, COCH₂), 2.35 (s, 3H, COCH₃),

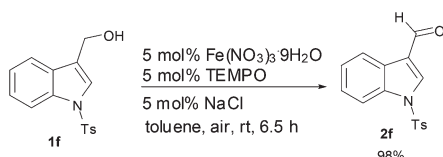
1.82–1.65 (m, 2H, CH₂), 1.43–1.25 (m, 4H, CH₂CH₂), 0.91 (t, J = 6.6 Hz, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 195.4, 144.8, 140.0, 138.4, 135.1, 129.4, 128.5, 127.4, 127.1, 124.2, 122.6, 115.9, 115.5, 42.5, 31.3, 24.3, 22.4, 21.5, 13.9; IR (neat) 1686, 1597, 1535, 1356, 1332, 1299, 1255, 1234, 1217, 1187, 1169, 1123, 1092, 1046 cm⁻¹; MS(EI) m/z (%) 369 (M⁺, 22.37), 91 (100); Anal. calcd for C₂₁H₂₃NO₃S: C 68.27; H 6.27; N 3.79; Found: C 68.19; H 6.13; N 3.51.

6. Synthesis of 1-tosyl-1H-indole-3-carbaldehyde (2f).

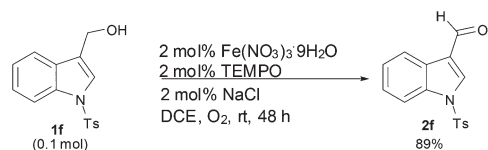


Method A: the reaction of (1-tosyl-1H-indol-3-yl)carbinol (301.3 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.6 mg, 0.1 mmol), TEMPO (15.8 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded 1-tosyl-1H-indole-3-carbaldehyde¹⁴ (238.4 mg, 80%) as a white solid (elute: petroleum ether–ethyl acetate = 5/1); m.p. 151–152 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 10.09 (s, 1H, CHO), 8.25 (d, J = 8.4 Hz, 1H, Ar-H), 8.23 (s, 1H, Ar-H), 7.95 (d, J = 7.5 Hz, 1H, Ar-H), 7.85 (d, J = 8.7 Hz, 2H, Ar-H), 7.45–7.32 (m, 2H, Ar-H), 7.28 (d, J = 8.7 Hz, 2H, Ar-H), 2.36 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 185.3, 146.1, 136.2, 135.1, 134.2, 130.3, 127.2, 126.24, 126.21, 125.0, 122.5, 122.3, 113.2, 21.6; IR (neat) 1662, 1595, 1538, 1479, 1444, 1399, 1372, 1279, 1237, 1168, 1126, 1098, 1082, 1016 cm⁻¹; MS(EI) m/z (%) 299 (M⁺, 49.19), 91 (100).



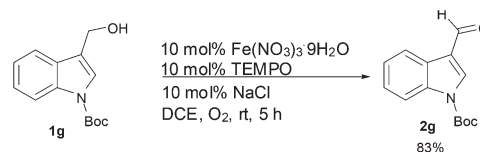
Method B: the reaction of (1-tosyl-1H-indol-3-yl)carbinol (301.8 mg, 1 mmol), Fe(NO₃)₃·9H₂O (20.8 mg, 0.05 mmol), TEMPO (7.9 mg, 0.05 mmol), and NaCl (2.9 mg, 0.05 mmol) in toluene (4 mL) afforded 1-tosyl-1H-indole-3-carbaldehyde (294.2 mg, 98%) as a white solid.



To a 250 mL three-necked flask were added Fe(NO₃)₃·9H₂O (0.8090 g, 2.0 mmol), TEMPO (0.3130 g, 2.0 mmol), NaCl (0.1180 g, 2.0 mmol) and DCE (100 mL). (1-tosyl-1H-indol-3-yl)carbinol (30.1984 g, 0.1 mol) was then added to the suspension and the resulting mixture was stirred at room temperature with an oxygen balloon until the reaction was complete, as monitored by TLC (elute: petroleum ether–ethyl acetate = 3/1). The resulting mixture was washed with saturated brine,

dried over anhydrous MgSO₄, concentrated *in vacuo* and recrystallized (DCM/petroleum ether) to afford 1-tosyl-1H-indole-3-carbaldehyde (26.5628 g, 89%) as a white solid.

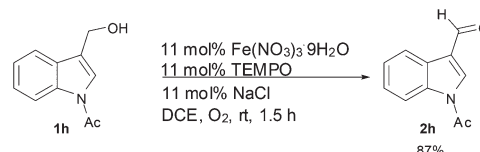
7. Synthesis of tert-butyl 3-formyl-1H-indole-1-carboxylate (2g).



Method A: the reaction of *tert*-butyl 3-(hydroxymethyl)-1H-indole-1-carboxylate (248.6 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.0 mg, 0.1 mmol), TEMPO (15.7 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded *tert*-butyl 3-formyl-1H-indole-1-carboxylate¹⁵ (204.0 mg, 83%) as a white solid: m.p. 129–130 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 10.10 (s, 1H, CHO), 8.29 (dd, J_1 = 6.9 Hz, J_2 = 1.8 Hz, 1H, Ar-H), 8.23 (s, 1H, Ar-H), 8.15 (d, J = 7.5 Hz, 1H, Ar-H), 7.45–7.34 (m, 2H, Ar-H), 1.71 (s, 9H, C(CH₃)₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 185.7, 148.8, 136.5, 135.9, 126.1, 126.0, 124.6, 122.1, 121.5, 115.1, 85.6, 28.0; IR (neat) 1740, 1675, 1556, 1482, 1449, 1396, 1355, 1327, 1310, 1273, 1238, 1153, 1131, 1099, 1044, 1016 cm⁻¹; MS(EI) m/z (%) 245 (M⁺, 12.63), 57 (100).

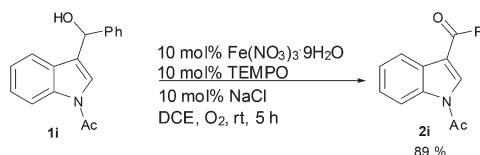
8. Synthesis of 1-acetyl-1H-indole-3-carbaldehyde (2h).



Method A: the reaction of 1-(3-(hydroxymethyl)-1H-indol-1-yl)ethanone (176.3 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.3 mg, 0.1 mmol), TEMPO (15.4 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded 1-acetyl-1H-indole-3-carbaldehyde¹⁶ (151.2 mg, 87%) as a pink solid (elute: petroleum ether–ethyl acetate = 3/1); m.p. 167–169 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 10.14 (s, 1H, CHO), 8.41 (d, J = 7.5 Hz, 1H, Ar-H), 8.28 (dd, J_1 = 6.6 Hz, J_2 = 1.8 Hz, 1H, Ar-H), 8.08 (s, 1H, Ar-H), 7.50–7.36 (m, 2H, Ar-H), 2.75 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 185.6, 168.5, 136.3, 135.1, 126.8, 126.0, 125.4, 122.6, 121.9, 116.3, 23.9; IR (neat) 1731, 1670, 1548, 1479, 1445, 1405, 1371, 1341, 1319, 1253, 1206, 1170, 1125, 1086, 1040, 1010 cm⁻¹; MS(EI) m/z (%) 187 (M⁺, 40.60), 144 (100).

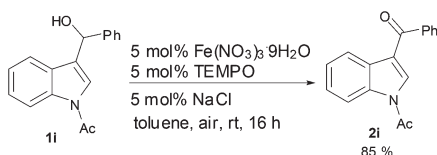
9. Synthesis of 1-(3-benzoyl-1H-indol-1-yl)ethanone (2i).



Method A: the reaction of 1-(3-(hydroxy(phenyl)methyl)-1H-indol-1-yl)ethanone (265.3 mg, 1 mmol), Fe(NO₃)₃·9H₂O

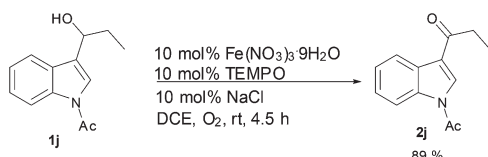
(40.0 mg, 0.1 mmol), TEMPO (15.7 mg, 0.1 mmol), and NaCl (5.8 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(3-benzoyl-1*H*-indol-1-yl)ethanone (233.3 mg, 89%) as a white solid (elute: petroleum ether–ethyl acetate = 5/1): m.p. 109–110 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.43 (d, *J* = 7.5 Hz, 1H, Ar-H), 8.26 (d, *J* = 7.2 Hz, 1H, Ar-H), 7.90–7.83 (m, 3H, Ar-H), 7.66–7.38 (m, 5H, Ar-H), 2.67 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 191.1, 168.6, 139.4, 136.0, 132.5, 132.2, 128.9, 128.6, 128.1, 126.4, 125.1, 122.4, 120.8, 116.2, 24.0; IR (neat) 1732, 1637, 1597, 1549, 1472, 1441, 1375, 1349, 1320, 1262, 1208, 1177, 1143, 1115, 1077, 1036, 1008 cm⁻¹; MS(EI) *m/z* (%) 263 (M⁺, 49.69), 144 (100); Anal. calcd for C₁₇H₁₃NO₂: C 77.55; H 4.98; N 5.32; Found: C 77.12; H 4.97; N 4.95.



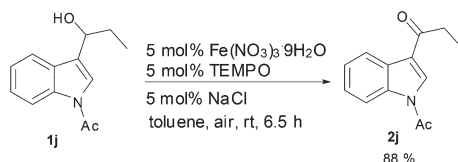
Method B: the reaction of 1-(3-(hydroxy(phenyl)methyl)-1*H*-indol-1-yl)ethanone (265.0 mg, 1 mmol), Fe(NO₃)₃·9H₂O (20.0 mg, 0.05 mmol), TEMPO (7.7 mg, 0.05 mmol), and NaCl (3.0 mg, 0.05 mmol) in toluene (4 mL) afforded 1-(3-benzoyl-1*H*-indol-1-yl)ethanone (223.9 mg, 85%) as a white solid.

10. Synthesis of 1-(1-acetyl-1*H*-indol-3-yl)propan-1-one (2j).



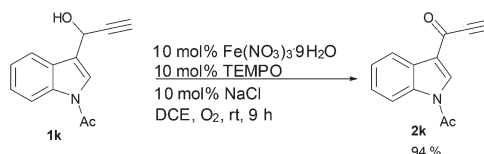
Method A: the reaction of 1-(3-(1-hydroxypropyl)-1*H*-indol-1-yl)ethanone (217.9 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.0 mg, 0.1 mmol), TEMPO (15.7 mg, 0.1 mmol), and NaCl (5.9 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(1-acetyl-1*H*-indol-3-yl)propan-1-one (192.1 mg, 89%) as a white solid (elute: petroleum ether–ethyl acetate = 5/1): m.p. 136–137 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.37–8.25 (m, 2H, Ar-H), 7.93 (s, 1H, Ar-H), 7.41–7.27 (m, 2H, Ar-H), 2.87 (q, *J* = 7.3 Hz, 2H, CH₂), 2.65 (s, 3H, COCH₃), 1.22 (t, *J* = 7.2 Hz, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 196.8, 168.6, 135.8, 130.3, 127.2, 126.0, 125.0, 122.3, 121.0, 116.0, 33.1, 23.9, 8.1; IR (neat) 1713, 1664, 1605, 1549, 1481, 1446, 1392, 1370, 1355, 1325, 1252, 1218, 1187, 1140, 1074, 1027 cm⁻¹; MS(EI) *m/z* (%) 215 (M⁺, 20.52), 144 (100); Anal. calcd for C₁₃H₁₃NO₂: C 72.54; H 6.09; N 6.51; Found: C 72.35; H 5.93; N 6.30.



Method B: the reaction of 1-(3-(1-hydroxypropyl)-1*H*-indol-1-yl)ethanone (207.0 mg, 1 mmol), Fe(NO₃)₃·9H₂O (20.6 mg, 0.05 mmol), TEMPO (7.6 mg, 0.05 mmol), and NaCl (2.9 mg, 0.05 mmol) in toluene (4 mL) afforded 1-(1-acetyl-1*H*-indol-3-yl)propan-1-one (180.7 mg, 88%) as a white solid.

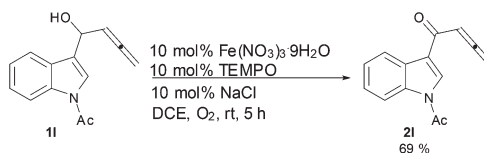
11. Synthesis of 1-(1-acetyl-1*H*-indol-3-yl)prop-2-yn-1-one (2k).



Method A: the reaction of 1-(3-(1-hydroxyprop-2-ynyl)-1*H*-indol-1-yl)ethanone (212.9 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.4 mg, 0.1 mmol), TEMPO (15.4 mg, 0.1 mmol), and NaCl (5.5 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(1-acetyl-1*H*-indol-3-yl)prop-2-yn-1-one (198.2 mg, 94%) as a white solid (elute: petroleum ether–ethyl acetate = 5/1): m.p. 151–152 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.39 (d, *J* = 7.5 Hz, 1H, Ar-H), 8.33 (d, *J* = 7.8 Hz, 1H, Ar-H), 8.25 (s, 1H, Ar-H), 7.48–7.35 (m, 2H, Ar-H), 3.31 (s, 1H, CH), 2.76 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 171.2, 168.5, 136.1, 135.1, 126.6, 126.0, 125.4, 122.3, 122.0, 116.2, 80.5, 77.4, 23.8; IR (neat) 2097, 1731, 1630, 1543, 1477, 1444, 1385, 1373, 1348, 1319, 1259, 1207, 1177, 1152, 1139, 1101, 1017 cm⁻¹; MS(EI) *m/z* (%) 211 (M⁺, 50.12), 169 (100); Anal. calcd for C₁₃H₉NO₂: C 73.92; H 4.29; N 6.63; Found: C 73.60; H 4.32; N 6.21.

12. Synthesis of 1-(1-acetyl-1*H*-indol-3-yl)butyl-2,3-dien-1-one (2l).



Method A: the reaction of 1-(1-acetyl-1*H*-indol-3-yl)butyl-2,3-dien-1-ol (227.0 mg, 1 mmol), Fe(NO₃)₃·9H₂O (40.0 mg, 0.1 mmol), TEMPO (15.7 mg, 0.1 mmol), and NaCl (5.7 mg, 0.1 mmol) in DCE (4 mL) afforded 1-(1-acetyl-1*H*-indol-3-yl)butyl-2,3-dien-1-one (154.7 mg, 69%) as a yellow solid: m.p. 126–128 °C (hexane–ethyl acetate).

¹H NMR (300 MHz, CDCl₃) δ 8.36 (d, *J* = 7.5 Hz, 1H, Ar-H), 8.33 (d, *J* = 7.5 Hz, 1H, Ar-H), 8.19 (s, 1H, Ar-H), 7.45–7.35 (m, 2H, Ar-H), 6.36 (t, *J* = 6.5 Hz, 1H, C=C=CH), 5.33 (d, *J* = 6.6 Hz, 2H, C=C=CH₂), 2.71 (s, 3H, CH₃); ¹³C NMR (75.4 MHz, CDCl₃) δ 215.7, 184.9, 168.5, 135.6, 130.9, 127.6, 126.2, 125.1, 122.4, 120.9, 116.0, 94.4, 79.5, 23.8; IR (neat) 1963, 1934, 1703, 1633, 1538, 1480, 1448, 1420, 1381, 1349, 1326, 1254, 1217, 1188, 1140, 1020 cm⁻¹; MS(EI) *m/z* (%) 225 (M⁺, 12.74), 144 (100); Anal. calcd for C₁₄H₁₁NO₂: C 74.65; H 4.92; N 6.22; Found: C 74.24; H 4.94; N 6.03.

Acknowledgements

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